

光電式轉速計 型號:DT-2234B

當你購買此台數位式轉速計時,你將踏入精密測試儀器的領域.雖然這台轉速計是一台複雜且精密的儀器.但如能正確使用的話.它將可以使用許多年.請詳細閱讀以下說明及注意事項.

1.特性

- *光電式轉速計.有很寬的測試範圍從 5 到 99999RPM.
- *0.1RPM 的解析度.測試值低於 1000RPM 時
- *記憶功能.最後值.最大值.最小值.自動儲存.可按 MEMORY 鍵逐一叫出讀值.
- *不足取的歸零抑制功能.使顯示器讀值得到正確的測量值.
- *容易讀取的液晶顯示器.能得到正確的 RPM 顯示值.避免不必要的猜測或錯誤.並且能節省電池能量.
- *這台轉速計.使用單一的微處理 LSI 積體電路.並以石英晶體做時間基準.所以可得到高準確的測試值.和快速的取樣時間.

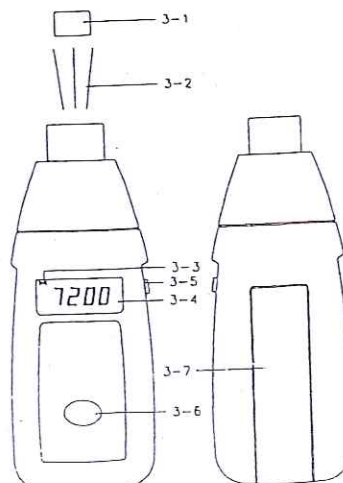
2.詳細規格說明

2-1.一般規格:

顯 示 器	5 位數.字高 10 公釐(0.4")液晶顯示器,和功能指示.
測 量 範 圍	5~99999RPM.
解 析 度	0.1RPM(<1000RPM). 1RPM(≥ 1000 RPM)
光 電 投 射 距 離	5~15 公分,如測量環境許可,最遠可達 30 公分.
精 確 度	$\pm(0.05\%+1 \text{ 位數})$.
取 樣 時 間	約 1 秒(超過 60rpm)
時 基	石英晶體振盪.4.194MHz.
線 路	單一超大型微處理積體電路.
電 池	1.5 伏特 3 號電池*4 顆.
工 作 環 境 溫 度	0~50°C (32~122°F).
工 作 環 境 濕 度	須低於 80%R.H.
外 觀 尺 寸	190*72*37mm.
重 量	235 公克.(包含電池)
標 準 附 件	攜帶手提盒.....1 只. 操作說明書.....1 本. 反光貼紙(60 公分長)....1 條.

3.面板說明:

- 3-1. 反射標誌(反光貼紙).
- 3-2. 光電式轉速測量投射光束.
- 3-3. 螢幕指示符號.
- 3-4. 液晶顯示器.
- 3-5. 測量按鍵.
- 3-6. 記憶值呼叫鍵.
- 3-7. 電池室/蓋..



4.轉速測量方法:

首先將反光貼紙剪一正方塊(至少須 $0.5\text{cm} \times 1.5\text{cm}$),貼於被測體上之外圍(須貼平),按下"測量按鍵"(圖 3-5),並使光束投射(圖 3-2)至被測體上之反光貼紙上.當光束投射至被測體時,可用"螢幕監視符號"(圖 3-3)來確任測量是否正確.當讀值穩定時(大約 2 秒鐘),即為被測體之轉速.此時可將放開測量按鍵.

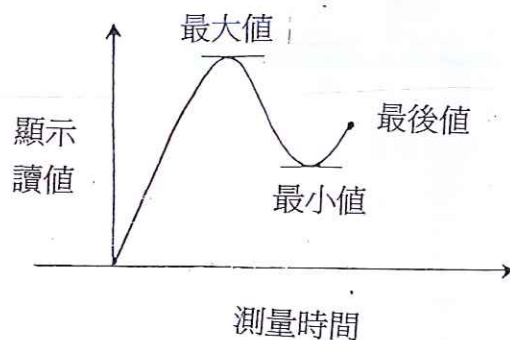
※如測量之轉速低時(低於 50RPM),建議將反光貼紙多貼一些(須平均分開),然後再進行測量,再將測得之轉速除以反射貼紙之數量,即可得到較高解析度和穩定的讀值.

記憶記錄功能之操作:

- A.當鬆開測量按鍵的同時.測量過程中的"最大值.最小值.最後值"將被自動地記憶下來.這些記憶值可使用"MEMORY"按鍵叫出.
- B.要呼叫出記憶儲存值.請依下列步驟操作:

須先釋放測量按鍵(圖 3-5).

- 1:按下記憶呼叫鍵一次.將會顯示出最後值.
顯示器將出現"LA"符號.
- 2:按下記憶呼叫鍵一次.將會顯示出最大值.
顯示器將出現"UP"符號.
- 3:按下記憶呼叫鍵一次.將會顯示出最小值.
顯示器將出現"dn"符號.



電池的更換:

- A.當顯示器出現"LO"符號時.即表示電池電力不足.必需更換新電池.
- B.打開電池蓋(圖 3-7).然後更換新電池.更換新電池後.再將電池蓋裝上.
- C.如長時間不使用此轉速計時.請將電池取出.

PHOTO TACHOMETER

Model : DT-2234B

ISO-9001, CE, IEC1010



LUTRON ELECTRONIC

The Art of Measurement

PHOTO TACHOMETER, Model : DT-2234B

CONTACT TACHOMETER, Model : DT-2235B

<i>Items</i>	PHOTO TACHOMETER			CONTACT TACHOMETER		
<i>Model</i>	DT-2234B			DT-2235B		
<i>Range</i>	5 to 99,999 RPM			0.5 to 19,999 RPM		
				<i>Surface Speed (m/min.) :</i> 0.05 to 1,999.9 m/min.		
				<i>Surface Speed (ft/min.) :</i> 0.2 to 6,560 ft/min.		
<i>Resolution</i>	RPM	0.1 RPM	< 1,000 RPM	RPM	0.1 RPM	< 1,000 RPM
		1 RPM	1,000 RPM		1 RPM	≥1,000 RPM
				Surface Speed	0.01 m/min.	< 100 m/min.
					0.1 m/min.	≥ 100 m/min.
					0.1 ft/min.	< 1000 ft/min.
					1 ft/min.	≥ 1,000 ft/min.
	<i>Display</i>			5 digits, 10 mm (0.4") LCD		
<i>Accuracy</i>			± (0.05 % + 1 digit).			
<i>Sampling Time</i>			1 second (60 RPM).			
<i>Time base</i>			Quartz crystal .			
<i>Photo Tach. detecting distance</i>	50 to 150 mm/2 to 6 inch. * Typical max. 300 mm/12 inch depending upon ambient light.					
<i>Circuit</i>	Exclusive one-chip of micro-computer LSI circuit.			Exclusive one-chip of micro-LSI circuit.		
<i>Battery</i>	1.5 V AA (UM-3) battery x 4 PCs.			1.5 V AA (UM-3) battery x 4 PCs.		
<i>Operating temp.</i>	0 to 50 °C (32 to 122 °F).			0 to 50 °C (32 to 122 °F).		
<i>Memory</i>	Last value, Max. value, Min. value.			Last value, Max. value, Min. value.		
<i>Size</i>	190 x 72 x 37 mm (7.5 x 2.8 x 1.5 inch).			208 x 72 x 37 mm (8.2 x 2.8 x 1.5 inch).		
<i>Weight</i>	250g (0.55 LB)/including battery.			280g (0.62 LB)/including battery.		
<i>Accessories</i>	Carrying case 1 PC Reflecting tape marks (600 mm) 1 PC Operation manual 1 PC			Carrying case 1 PC RPM adapter (CONE)..... 1 PC RPM adapter (FUNNEL)..... 1 PC Surface speed test wheel..... 1 PC Operation manual 1 PC		

* Appearance and specification listed in this brochure are subject to change without notice.

0601-DT2234B/DT2235B

A decorative border made of a laurel wreath, consisting of two branches of leaves and berries that curve upwards at the top and downwards at the bottom, framing the central text.

**PHOTO
TACHOMETER**

**OPERATION
MANUAL**



PHOTO TACHOMETER

Your purchase of this PHOTO TACHOMETER marks a step forward for you into the field of precision measurement. Although this TACHOMETER is a complex and delicate instrument, its durable structure will allow many years of use if proper operating techniques are developed. Please read the following instructions carefully and always keep this manual within easy reach.

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1. FEATURES

- * Measuring RPM is safe & accurate without attachment to object.
- * Wide measuring range & high resolution.
- * Digital display gives exact RPM with no guessing or errors.
- * Used the exclusive MICRO-COMPUTER LSI-circuit and crystal time base to offer the high accuracy measurement & fast measuring time.
- * The last value/max. value/min. value will be automatically stored in memory and can be obtained by pressing MEMORY CALL BUTTON.
- * The use of durable, long lasting components, including a strong, light weight ABS-plastic housing, assures maintenance free performance for many years. The housing has been carefully shaped to fit comfortably in either hand.

2. SPECIFICATIONS

Display	5 digits, 10 mm (0.4 ") LCD (Liquid Crystal Display), with function annunciation.
Measurement Range	5 to 99,999 RPM
Resolution	0.1 RPM (5 to 999.9 RPM) 1 RPM ($\geq$ 1,000 RPM)

Accuracy (23 ± 5 °C)	± (0.05 % + 1 digit). * Spec. tested under the environment RF field strength less than 3 V/M and frequency less than 30 MHz only.
Time base	Quartz crystal, 4.194 MHz.
Circuit	Exclusive one-chip of microcomputer LSI circuit.
Sampling Time	1 sec. (over 60 RPM).
Test range select	Automatic.
Memory	Last, Maximum, Minimum value.
Detecting Distance	50 to 150 mm/2 to 6 inch (typical max. 300 mm/12 inch, depending upon ambient light).
Battery	4 x 1.5V AA (UM-3) battery.
Power Consumption	Approx. DC 52 mA.
Operating Temperature	0 - 50 °C (32 - 122 °F)
Size	200 x 77 x 37 mm (7.5 x 3.0 x 1.5 inch)
Weight	185 g (0.41 LB)
Accessories Included	Carrying case.....1 PC Reflecting tape marks (600 mm)....1 PC Operation manual.....1 PC

3. FRONT PANEL DESCRIPTION

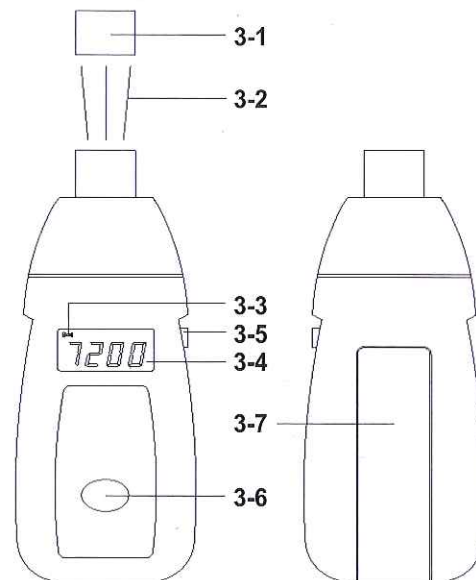


Fig. 1

Fig. 1

- 3-1 Reflective Mark
- 3-2 Signal light Beam
- 3-3 Monitor indicator
- 3-4 Display
- 3-5 Measuring Button
- 3-6 Memory Call Button
- 3-7 Battery Compartment/Cover

4. MEASURING PROCEDURE

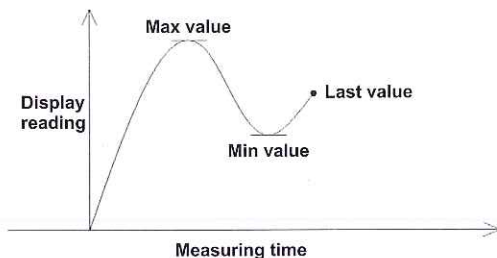
- 1) Apply a reflective mark to the object being measured.
- 2) Depress the MEASURING BUTTON (3-5, Fig. 1) and align the visible light beam (3-2, Fig. 1) with the applied target. Verify that the MONITOR INDICATOR (3-3, Fig. 1) lights when the target pass through the stabilizes (about 2 seconds).

Note :

If the measured RPM values is very low, for example less than 50 RPM, recommend to attach more " Reflecting Marks " to the object average. It will get the real RPM with high resolution, precisely & fast sampling time when divided the reading values by the no. of the " Marks ".

5. MEMORY CALL BUTTON OPERATION

- 1) The readout of " last value ", " max. value " & " min. value " can be obtained immediately & memorized into the circuit automatically before turning off the " Measuring Button " (3-6, Fig. 1)



- 2) When finish the measuring procedures (after release the measuring button), the memorized values can be displayed on the LCD display whenever :
 - a) First push the " Memory Call Button " (3-6, Fig. 1) to display the last value (" LA " and the " last value " will be displayed alternately).
 - b) Second push the " Memory Call Button " (3-6, Fig. 1) to display the maximum value (" UP " and the " max. value " will be displayed alternately).
 - c) Third push the " Memory Call Button " (3-6, Fig. 1) to display the minimum value (" dn " and the " min. value " will be displayed alternately).

6. BATTERY REPLACEMENT

- 1) When it is necessary to replace the battery (battery voltage less than approx. 4.5 V), the " LO " text will appear on the display.
- 2) Slide the battery cover (3-7, Fig. 1) away from the the instrument and remove the battery.
- 3) Install the batteries (4 x 1.5V AA/UM-3) correctly into the case. Permanent damage to the circuit may result from incorrect installation.